

PRD.md

Loadster - Distributed Load Testing Platform

Vision

Loadster is a modern cloud-native load testing platform inspired by k6, JMeter, and Locust.

The project provides:

- Local CLI load testing
- Distributed Kubernetes-based load testing
- Prometheus metrics export
- Grafana dashboard integration
- Scenario-based load testing
- Cloud-native deployment support

The goal is to simulate realistic traffic patterns against HTTP services while collecting detailed performance metrics.

Target Users

1. Backend Engineers
 2. DevOps Engineers
 3. SRE Engineers
 4. Platform Engineers
 5. Students learning cloud-native systems
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Core Features

Feature 1: CLI Load Testing

Example:

```
loadster run
--url https://api.example.com
--users 100
--duration 60s
```

Expected Output:

- Requests Sent
- Requests Failed

- Success Rate
 - Average Latency
 - P50
 - P90
 - P95
 - P99
 - Requests Per Second
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Feature 2: Scenario Testing

Support YAML test definitions.

Example:

target: https://api.example.com

stages:

- users: 10 duration: 30s
- users: 100 duration: 1m
- users: 500 duration: 2m

The engine should automatically ramp traffic.

Feature 3: Live Terminal Dashboard

Display:

- Active Users
- Current RPS
- Average Latency
- Error Count
- Stage Progress

Updated in real time.

Feature 4: Metrics Export

Expose:

/metrics

Prometheus must be able to scrape:

- requests_total
 - failed_requests_total
 - latency_histogram
 - rps_current
 - active_users
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Feature 5: Distributed Mode

Example:

loadster distributed run scenario.yaml

Master node coordinates:

- Worker Pods
- Scenario Distribution
- Result Aggregation

Workers generate traffic.

Feature 6: Report Generation

Generate:

- JSON reports
- HTML reports

Include:

- Throughput
 - Error Rate
 - Latency Distribution
 - Test Duration
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Non Functional Requirements

- Cross Platform
- Linux
- macOS
- Windows

Performance:

- Support 10,000+ virtual users

- Memory Efficient
- Low CPU Overhead

Reliability:

- Graceful Shutdown
- Retry Logic
- Fault Tolerance

Security:

- TLS Support
 - Secure Configuration
 - No Hardcoded Secrets
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Success Metrics

- Generate 10k+ virtual users
- Prometheus integration works
- Distributed Kubernetes mode operational
- Full CI/CD pipeline operational
- Helm deployment available